

 CYBERSECURITY TECH PRESENTATION

Steganography vs Cryptography

Hidden Security & Data Concealment Techniques in the Digital World



Department of Computer Science & Engineering | B.Tech Presentation

Introduction to Data Hiding

-  **Data Protection Need:** Rapid digitalization requires robust safeguards to defend sensitive communications across untrusted public networks.
-  **Encryption Paradigm:** Cryptography scrambles plaintext into ciphertexts to keep message content secret from unauthorized interceptors.
-  **Concealment Paradigm:** Steganography hides the very existence of a communication channel by embedding payloads inside ordinary cover media.
-  **Complementary Security:** Combining both techniques yields multi-layered defense-in-depth, protecting both context and content.



Core Terminology & Definitions

Cryptography ("Secret Writing")

Derived from Greek *kryptos* (hidden) + *graphein* (writing).

- ▷ **Definition:** Art & science of transforming readable plaintext into unreadable ciphertext using mathematical keys.
- ▷ **Primary Focus:** Securing data content against unauthorized access or tampering.
- ▷ **Visibility:** Highly conspicuous; alerts attackers that secret information is being sent.
- ▷ **Core Standard:** AES, RSA, ECC, SHA-256.

Steganography ("Covered Writing")

Derived from Greek *steganos* (covered) + *graphein* (writing).

- ▷ **Definition:** Concealing secret data inside innocuous cover files (images, audio, video, text) without noticeable distortion.
- ▷ **Primary Focus:** Concealing the existence of the communication.
- ▷ **Visibility:** Inconspicuous; appears as standard, harmless multimedia traffic.
- ▷ **Core Standard:** LSB Insertion, DCT, Masking.

Working Mechanisms

LSB & Mathematical Encryption

Cryptographic Flow: Plaintext undergoes complex mathematical permutation/substitution driven by a key, producing a high-entropy ciphertext block.

LSB Steganography Flow: Replaces the Least Significant Bit of binary color bytes (RGB) in an image with payload bits.

Imperceptibility Factor: Modifying the LSB alters a pixel color value by at most 1 unit (out of 255), keeping visual alterations completely invisible to the human eye.

Comparative Analysis Matrix

Comparison Attribute	Cryptography	Steganography
Primary Objective	Hide the <i>meaning</i> / <i>content</i> of data	Hide the <i>existence</i> of communication
Data Appearance	Scrambled, unreadable ciphertext	Innocuous cover object (Image/Audio/Video)
Visibility Level	High (attracts interceptor attention)	Low (undetected by casual observers)
Security Principles	Confidentiality, Integrity, Non-repudiation	Confidentiality, Stealth/Concealment
Adversarial Attack	Cryptanalysis (Mathematical cracking)	Steganalysis (Statistical anomaly detection)
Key Dependence	Mandatory cryptographic key exchange	Optional stego-key for embedding control

✓ Key Advantages & Synergies



Cryptography Strengths

- High mathematical security backed by hard algorithms.
- Guarantees authentication via digital signatures.
- Ensures data integrity with hash verifications.



Steganography Strengths

- Zero suspicion avoids targeted intercept attacks.
- Passes through firewall filters unflagged.
- Ideal for covert ops in censored regimes.



Crypto-Stegano Hybrid

- First encrypt payload using AES-256 cipher.
- Embed ciphertext into LSB of cover image.
- Offers robust two-tier defense-in-depth security.

⚠ Limitations & Vulnerabilities

🔒 Cryptography Limitations

- **Target Flagging:** Ciphertexts immediately arouse suspicious interest from network monitors.
- **Computational Cost:** Public-key encryption (RSA/ECC) incurs high CPU latency overhead.
- **Key Management:** Compromise or loss of private keys leads to total system failure.

👁 Steganography Limitations

- **Payload Constraints:** Large payloads degrade cover quality, revealing hidden artifacts.
- **Fragility to Processing:** Image compression (JPEG), cropping, or resizing destroys payload.
- **Steganalysis Sensitivity:** Statistical Chi-square tests can reveal bit anomalies.

Real-World Applications

-  **Digital Watermarking:** Embedding invisible copyright ownership markers inside digital images and audio files.
-  **Banking & E-Commerce:** TLS/SSL encryption and AES payload encryption for safe financial transactions.
-  **Medical Data Privacy:** Hiding patient EHR history inside DICOM medical imaging files to meet privacy laws.
-  **Cyber Intelligence:** Threat actors & defenders utilizing hidden C2 channels and covert log storage.



Security & Attack Vector Analysis

2-Tier

Hybrid Security Architecture

Combining AES Encryption with LSB Steganography ensures secrecy even if cover media is compromised.

-  **Cryptanalysis:** Focuses on key recovery, differential cryptanalysis, and mathematical vulnerabilities in cipher implementations.
-  **Steganalysis:** Employs high-order statistical analysis, histograms, and deep learning classifiers to detect hidden payloads.
-  **The Security Trade-off:** Steganography trades capacity for stealth, whereas Cryptography trades stealth for robust mathematical proof.

Conclusion & Summary

"Cryptography protects the content of the message, while Steganography protects the context and existence of the message."

■ ■ Key Academic References

- W. Stallings, *Cryptography and Network Security: Principles and Practice*, Pearson.
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